

Separable Differential Equations

Lecture Notes

Topics: Separable Form · Solution Method · Exponential Growth · Newton's Cooling · Falling Golf Ball

Where We Left Off

Last class we asked: *what is a differential equation, and what can we say about a system without solving it?* We could:

- Use sign analysis to find equilibria and predict long-run behavior.
- Sketch the slope field and trace trajectories.
- Verify whether a candidate formula satisfies a given DE.

Today we move from *checking* a solution to *producing* one. We'll focus on a particular class of first-order ODEs where the algebra works out cleanly.

Note

There are many types of differential equations and many tricks to solve them. Like integrals, some DEs do not have closed form solutions. Today we handle a common form of differential equation. In future courses, you'll see more complex versions.

Separable Equations

Separable First-Order ODE

A first-order ODE

$$\frac{dy}{dx} = f(x, y)$$

is **separable** if the right-hand side can be written as a product of a function of x alone and a function of y alone:

$$\frac{dy}{dx} = f(x) g(y).$$

This means that we can isolate all the y 's on one side and all the x 's on the other:

$$\frac{dy}{g(y)} = f(x) dx.$$

The Solution Method

Solving a separable ODE

Given $\frac{dy}{dx} = f(x) g(y)$:

1. **Separate** variables: $\frac{dy}{g(y)} = f(x) dx$.

2. **Integrate** both sides: $\int \frac{dy}{g(y)} = \int f(x) dx$.
3. **Solve for y** as a function of x . Use the initial condition to pin down the constant of integration.

Note

Why is splitting up dy and dx allowed? It looks like we're treating $\frac{dy}{dx}$ as a fraction, which is not quite right. The shorthand works because of u -substitution. Here is the careful version:

- (1) Start from $\frac{1}{g(y)} \frac{dy}{dx} = f(x)$.
- (2) Integrate both sides with respect to x : $\int \frac{1}{g(y)} \frac{dy}{dx} dx = \int f(x) dx$.
- (3) On the left, u -substitution with $u = y(x)$, $du = \frac{dy}{dx} dx$ turns the integrand into $\frac{du}{g(u)}$, giving $\int \frac{du}{g(u)} = \int f(x) dx$.

Same answer either way. The “separate the differentials” move just skips writing out the u -sub.

Example 1: Exponential Growth

$$\frac{dy}{dt} = k y, \quad y(0) = y_0$$

Setup: the rate of change is proportional to the current value, with constant of proportionality k . When $k > 0$ this is exponential growth (faster growth when y is larger); when $k < 0$ it's exponential decay.

Step 1: Separate.

$$\frac{dy}{y} = k dt.$$

Step 2: Integrate both sides.

$$\int \frac{dy}{y} = \int k dt \implies \ln |y| = k t + C.$$

Step 3: Solve for y . Exponentiate both sides:

$$|y| = e^{kt+C} = e^C e^{kt}.$$

Letting $A = \pm e^C$ (a new arbitrary constant) absorbs both the sign and the e^C :

$$y(t) = A e^{kt}.$$

Apply the initial condition: $y(0) = A e^0 = A$, so $A = y_0$.

Answer:

$$y(t) = y_0 e^{kt}$$

Sanity check: differentiate: $y'(t) = k y_0 e^{kt} = k y(t)$. ✓

Note

This formula should look familiar: it's the same shape as the present-value relationship $PV = C e^{-rT}$ we used in Block 2 for continuous discounting. $C e^{-kt}$ is exponential decay; $C e^{kt}$ is exponential growth. The same mathematical structure shows up across populations, radioactive decay, drug clearance, and continuously compounded interest.

Example 2: Newton's Law of Cooling

Last class we *verified* that $T(t) = 22 + 68 e^{-kt}$ solves the cooling equation. Today we *derive* it.

$$\frac{dT}{dt} = -k(T - T_a), \quad T(0) = T_0$$

Setup: with $T_a = 22^\circ\text{C}$ and $T_0 = 90^\circ\text{C}$. Try to separate directly: $\frac{dT}{T - T_a} = -k dt$. We could integrate this as-is, but a substitution makes the structure even cleaner and reveals why this equation behaves just like exponential decay.

Substitution: let

$$u(t) = T(t) - T_a.$$

Since T_a is constant,

$$\frac{du}{dt} = \frac{dT}{dt} = -k(T - T_a) = -k u.$$

This is exactly the exponential-decay equation from Example 1 with growth rate $-k$.

Apply Example 1: the solution to $\frac{du}{dt} = -k u$ is

$$u(t) = u_0 e^{-kt} \quad \text{where} \quad u_0 = u(0) = T_0 - T_a.$$

Unwind the substitution: replace u by $T - T_a$:

$$T(t) - T_a = (T_0 - T_a) e^{-kt} \implies T(t) = T_a + (T_0 - T_a) e^{-kt}$$

With our numbers, $T_a = 22$ and $T_0 = 90$, so $T(t) = 22 + 68 e^{-kt}$, matching what we verified last class.

Sanity check:

- $T(0) = 22 + 68 e^0 = 22 + 68 = 90$. ✓
- As $t \rightarrow \infty$, $e^{-kt} \rightarrow 0$, so $T(t) \rightarrow 22 = T_a$. ✓ (Coffee cools to room temperature.)

Note

Where does k come from? The rate constant k depends on the physical setup (mug, room airflow, coffee volume). In practice you find k from data: take temperature readings over time, take $\ln(T - T_a)$, fit a line by linear regression, and read off the slope (which equals $-k$). With $k = 0.05 \text{ min}^{-1}$ the coffee in our example takes roughly 90 minutes to come close to room temperature.

Example 3: The Falling Golf Ball

Returning to the equation we set up last class: $\frac{dv}{dt} = g - kv$, with $g = 9.8 \text{ m/s}^2$, $k = 0.31 \text{ s}^{-1}$, and $v(0) = 0$ (dropped from rest).

$$\frac{dv}{dt} = g - kv, \quad v(0) = 0$$

Setup: we could separate directly, but the same substitution trick as in Example 2 works beautifully here. The equilibrium velocity is $v = g/k$. Measure v relative to the equilibrium.

Substitution: let

$$u(t) = v(t) - \frac{g}{k}.$$

Since g/k is constant,

$$\frac{du}{dt} = \frac{dv}{dt} = g - kv = g - k\left(u + \frac{g}{k}\right) = g - ku - g = -ku.$$

Same exponential-decay equation again.

Apply Example 1:

$$u(t) = u_0 e^{-kt} \quad \text{where} \quad u_0 = u(0) = 0 - \frac{g}{k} = -\frac{g}{k}.$$

Unwind the substitution:

$$v(t) - \frac{g}{k} = -\frac{g}{k} e^{-kt} \implies \boxed{v(t) = \frac{g}{k} (1 - e^{-kt})}$$

Sanity check:

- $v(0) = \frac{g}{k}(1 - 1) = 0$. ✓ (Dropped from rest.)
- As $t \rightarrow \infty$, $e^{-kt} \rightarrow 0$, so $v(t) \rightarrow \frac{g}{k}$. ✓ (Approaches terminal velocity.)

Exponential Growth and Decay

In all three examples the same equation kept reappearing:

$$\frac{du}{dt} = -ku, \quad \text{with solution} \quad u(t) = u_0 e^{-kt}.$$

For Newton's cooling and the falling golf ball, a constant shift turned the original equation into this canonical form. This shape ($u' = -ku$) is the workhorse of first-order linear dynamics: it shows up in radioactive decay, drug clearance, and RC circuit charging.